GLOBAL TRADE DOMINANCE

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Abstract

In 2000, the USA and the EU dominated trade with much of the word.By 2024, however, China has overtaken both to become the largest trading partner for nearly all of Asia,Africa, and South America. This shift illustrate the dynamic nature of global trade dominance, defined as the ability of a country to influence international market through exports, imports, competitiveness, and economic resilience. This project investigates the evolution of trade performance among leading economies using data from the World Bank Global Economic Monitor covering the period from 2006-2024. Key variables include export, import of merchandise (constant US$), nominal and real effective exchange rates, consumer price indices, and total reserves. After appropriate data cleaning and transformation, countries were first grouped into clusters based on their trade and macroeconomic profiles. From these clusters, representative economies Ghana from the emerging economies cluster and Brazil from the more advanced economies cluster were selected for deeper forecasting analysis using time-series models.
The results highlight diverging patterns in trade resilience and competitiveness across economies. These findings underscore the shifting balance of trade power and provide valuable insights for policymakers, investors, and economists concerned with the future dynamics of global trade.

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1 Introduction

Global trade is one of the central forces shaping economic growth, geopolitical relations, and financial stability. The notion of *trade dominance* refers to the ability of a country to exert influence over international markets by means of its trade performance, competitiveness, and resilience to external shocks. Such dominance is shaped by multiple economic factors, including merchandise exports and imports, terms of trade, exchange rate dynamics, price stability, industrial production, and foreign reserves. Examining the interplay of these variables is essential to understand not only current patterns but also the potential trajectories of global economies.

While past analyses of trade often focus on descriptive comparisons between countries, less attention has been given to forward-looking projections that anticipate shifts in dominance. Forecasting methods can provide valuable insights into the sustainability of current positions and the possible emergence of new trade powers.

This project therefore seeks to answer the following research question: *How will economies sustain or lose their trade dominance in the next decade, and which economic factors will prove decisive in shaping these outcomes?*

To address this question, the study combines clustering techniques with time series forecasting. Countries are first grouped into clusters according to their macroeconomic and trade structures, after which two representative economies are selected for deeper analysis: Brazil, representing a large emerging economy with diversified trade and resource-based strength, and Ghana, representing a fast growing economy in Sub-Saharan Africa with high exposure to structural challenges.

1.1 Data sources

- **The Global Economic Monitor (GEM)[3]** , curated by the World Bank, provides high-frequency economic indicators designed to track the short- and medium-term evolution of economies worldwide. It consolidates data from national authorities, international organizations, and World Bank country teams, making it one of the most widely used references for international trade and macroeconomic research. The GEM database contains monthly, quarterly, and annual indicators for over 200 countries and regions. The dataset spans from the early 1960s to the present, depending on data availability, and includes both seasonally adjusted and unadjusted series. Its broad coverage enables consistent cross-country comparisons and timely monitoring of global economic developments.

  The indicators are grouped into thematic categories, including:

  - **Real Sector:** Industrial production, retail sales, and output indices.
  - **External Sector:** Merchandise exports and imports, trade balance, and terms of trade.
  - **Prices and Exchange Rates:** Consumer price indices, nominal and real effective exchange rates, and commodity prices.
  - **Monetary and Financial Indicators:** Total reserves and interest rates.

- **Global Economy Indicators**

  To complete the GEM dataset, we also used the Kaggle Global Economy Indicators[1] which is compiled by the National Accounts Main Aggregates Database[2].This dataset aggregates annual macroeconomic indicators for over 180 countries, covering GDP, trade flows, inflation, foreign direct investments, reserves, unemployment and more. It's broad coverage allows for cross sectional comparisons between countries, making it suitable for clustering analysis. By grouping countries with

similar trade and macroeconomics profiles, this dataset provides a basis for selecting representative economies for each cluster to perform forecasting on the GEM time series data.

The integration of these two datasets enables both breadth and depth of analysis.

- The Kaggle dataset captures structural differences across countries, allowing us to identify clusters of economies with similar trade patterns.

- The GEM dataset provides high frequency time series data for selected representative countries, which are then used for forecasting exports and trade balance.

1.2 Selected Variables

For the clustering, a set of variable was selected from the Global Economy Indicators dataset to get more informations about each countries macroeconomics. The table 1 below summerizes the indicators included in the analysis.

Variable	**Description**
exports	Exports of goods and services (% of GDP)
imports	Imports of goods and services (% of GDP)
gdp	Gross Domestic Product (constant 2015 US$)
gni_usd	Gross National Income (constant 2015 US$)
s_total_va	Total value added (% of GDP)
s_agri	Agriculture, forestry, and fishing value added (% of total value added)
manufact	Manufacturing, mining, and utilities value added (% of total value added)
mining_manuf_utils	Mining and manufacturing value added (% of total value added)
services_trade	Services trade (% of GDP)
transport	Transport, storage, and communication value added (% of total value added)
household_consumption	Household final consumption expenditure (% of GDP)
gov_consumption	General government final consumption expenditure (% of GDP)
s_gfcf	Gross fixed capital formation (% of GDP)
s_pop	Total population
ama	Exchange rate

Table 1: Selected Global Economic Indicators variables used in the analysis.

A subset of GEM variables with a percentage of missing values below 35% was selected to capture the notion of *trade dominance*. The table 2 summarizes the indicators included in the analysis.

Variable	Description
Imports_USD_SeasonalAdj (Thousands)	Merchandise imports, seasonally adjusted, in constant US$ (thousands).
Exports_USD_SeasonalAdj (Thousands)	Merchandise exports, seasonally adjusted, in constant US$ (thousands).
Trade Balance	Derived variable: Exports minus Imports.
CPI_Nominal_SeasonalAdj	Consumer Price Index, nominal values, seasonally adjusted.
CPI_Nominal_NoSeasonalAdj	Consumer Price Index, nominal values, not seasonally adjusted.
CPI_YoY_Nominal_SeasonalAdj	Year-on-year change in nominal Consumer Price Index, seasonally adjusted.
ExchangeRate_Official_PeriodAvg	Official exchange rate, period average.
Total_Reserves (Thousands)	Total international reserves, reported in constant US$ (thousands).
ForeignReserves_ImportCover_Months	Ratio of foreign reserves to monthly imports, expressed in months of coverage.

Table 2: Selected GEM variables used in the analysis.

1.3 Descriptive Statistics

To provide an overview of the dataset, Table 3 reports summary statistics for variables:

Variable	Mean	Std. Dev.	Min	Max
Imports_USD_SeasonalAdj (Thousands)	77.51	319.03	0.01	3619.57
Exports_USD_SeasonalAdj (Thousands)	42.93	172.25	0.00	1972.78
Trade Balance	-34.58	183.04	-1646.80	133.65
CPI_Nominal_SeasonalAdj	142.13	242.66	58.13	8919.95
CPI_Nominal_NoSeasonalAdj	146.94	259.64	58.11	8933.98
CPI_YoY_Nominal_SeasonalAdj	5.88	11.20	-6.52	266.41
ExchangeRate_Official_PeriodAvg	1036.15	7604.82	0.21	245038.30
Total_Reserves (Thousands)	362.02	1387.28	0.03	14485.98
ForeignReserves_ImportCover_Months	7.24	8.41	0.00	132.40

Table 3: Descriptive statistics of selected GEM variables (2006–2024).

1.4 Relevance for This Study

By focusing on these variables, the dataset provides the empirical basis for analyzing trade resilience and forecasting the future performance of dominant economies. These indicators capture the interaction between trade flows, price dynamics, monetary stability, and reserve adequacy, key components of global trade dominance.

2 Methodologies

2.1 Data preprocessing for Global Economic Indicators

The first major choice was to transform the data from a long format (country, year, variable, value) into a wide format where each country represents a row an the economics variables and years are columns, which was crucial for preparing the data for a clustering algorithm, which typically operates on a matrix of observations (countries) and features (economic indicators).

2.1.1 Feature Construction

The raw dataset contains dozens of time-series variables with heterogeneous scales, units, and direct economic meanings. For this reason, we implemented a feature-extraction function in Python (compute_country_features that converts raw time series into economically interpretable summary statistics. These features are comparable across countries and directly reflect economic concepts (growth, volatility, sectoral structure, etc.).
For readability, we present only representative examples of the constructed features; the complete list of features is reported in the Appendix.

Table 4: Constructed feature groups, representative examples, and interpretation

Feature group	**Representative features**	**Economic interpretation**
Scale	- `gdp_latest`: latest GDP level - `gni_percap_latest`: GNI per capita - `population_latest`: latest population	Captures country size and income level; distinguishes microstates from large economies and advanced from developing countries.
Dynamics	-`gdp_mean_growth` = mean $\Delta\log(GDP_t)$ -`gdp_vol_growth`=std($\Delta\log(GDP_t)$) -`exp_vol_growth`=std($\Delta\log(Exports_t)$) -`exp_max_drawdown`=$\min_t(\log Exports_t - \max_{s\leq t}\log Exports_s)$	Summarizes stability and resilience: identifies countries with steady growth vs volatile, commodity-driven or crisis-prone paths.
Structure	-`manufacturing_share_va`= VA_{manuf}/VA_{total} -`agri_share_va` = VA_{agri}/VA_{total} -`sector_HHI`=$\sum_i s_i^2$	Captures productive composition and diversification; high HHI indicates specialization, low HHI diversification.
External linkages	- `exports_share_gdp` = $Exports/GDP$ - `imports_share_gdp` = $Imports/GDP$ -`trade_openness`= $(Exports+Imports)/GDP$ - `corr_exp_gdp` = corr($\Delta\log Exports$, $\Delta\log GDP$)	Measures trade dependence and sensitivity of domestic output to external demand.
Exchange rate	- `fx_mean`: average log exchange rate - `fx_vol` = std($\Delta\log(FX_t)$) - `fx_trend`: slope of $\log FX_t$ - `fx_ac1`: lag-1 autocorrelation	Indicates stability or volatility of the currency, risks of depreciation/appreciation, and exposure to financial shocks.

2.1.2 Missing Values Management and Feature Selection

The preprocessing pipeline involved several key decisions aimed at creating a robust and clean dataset for clustering. This process focused on handling missing values, mitigating the effect of extreme outliers, and

reducing feature redundancy.

Outlier Mitigation with Winsorizing: A primary concern was the presence of extreme outliers within the dataset. These values can disproportionately influence distance-based clustering algorithms like K-Means by pulling cluster centroids towards them. To address this, we applied **Winsorizing**, a method that caps extreme values at a specific percentile, in this case, the 1st and 99th percentiles. This approach was chosen to:

- **Limit Distortion:** Outliers can distort the distance between data points, pulling the centroid of a cluster in an incorrect direction. Winsorizing prevents this by limiting the influence of extreme values, making the clusters more robust.
- **Enhance Robustness:** Many clustering algorithms, including K-Means, rely on Euclidean distance calculations. An extreme value in a single dimension can artificially inflate these distances, leading to less meaningful groupings. By reducing the spread of these values, Winsorizing allows the algorithm to focus on the patterns within the majority of the data.

Imputation and Scaling: Following outlier mitigation, missing values were addressed using a **K-Nearest Neighbors (KNN) Imputer** (KNNImputer, scikit-learn). We set the number of neighbors to k=5, a conventional compromise: fewer neighbors would risk noisy imputations driven by single outliers, while more neighbors would blur country-specific patterns. This method fills in missing values based on the data from the most similar countries. Tests with k=3 and k=7 produced similar results, confirming robustness. This is a more complete approach than simple mean or median imputation, as it preserves the relationships between countries within the feature space.
For feature scaling, we used **RobustScaler** instead of the more common `StandardScaler`. The key distinction is that while `StandardScaler` uses the mean and standard deviation (which are sensitive to outliers), `RobustScaler` utilizes the median and the interquartile range (IQR). This choice provides two main benefits:

- **Resistance to Outliers:** Even after Winsorizing, some residual outliers may exist. RobustScaler offers an additional layer of protection, ensuring the scaling process itself is not distorted.
- **Fair Weighting of Variables:** For distance-based clustering algorithms, it is crucial that all variables contribute equally. `RobustScaler` ensures that variables with larger value ranges do not dominate the distance calculations.

Feature Reduction The final step was to reduce feature redundancy by removing highly correlated variables. We established a threshold of 0.85 for the Pearson correlation coefficient. Any pair of variables with a correlation above this threshold had one of the variables removed. The rationale behind this decision was to:

- **Avoid Redundancy:** Highly correlated features provide redundant information and can give undue weight to certain economic concepts during clustering.
- **Improve Model Performance and Interpretability:** By simplifying the feature set, we can improve the performance of the clustering algorithm and make the resulting clusters easier to interpret and describe.

The dropped features were the following:'gni_percap_latest', 'trade_openness', 'va_mean_growth', 'va_vol_growth' 'primary_share', 'manufacturing_share'.

2.2 Data Preprocessing for GEM dataset

Before applying forecasting techniques, the dataset was cleaned. Missing values were handled through interpolation when feasible, while irrelevant or inconsistent entries were excluded. Derived variables, such as the trade balance (exports minus imports), were constructed to capture core trade dynamics. Seasonal adjustments were already provided by the GEM dataset for key variables, ensuring comparability across time and reducing cyclical noise.

2.2.1 Missing Values Management in the GEM Dataset

A different approach was employed to manage missing values. Given the nature of this dataset and its variables, we chose to use a machine learning-based imputation model called **IterativeImputer**.

The IterativeImputer Model Unlike simpler methods like mean or median imputation, which fill missing values one column at a time without considering relationships between features, `IterativeImputer` is a more sophisticated model. It works by treating each feature with missing values as a target variable to be predicted based on the other features in the dataset.

The process is as follows:

1. Initially, a simple imputation (e.g., using the mean) is performed to fill all missing values as a starting point.
2. The model then iterates through each feature that has missing values. For a given feature, all its known values are used along with the other features to train a regression model (random forest).
3. This trained model is then used to predict and fill in the missing values for that specific feature.
4. This process is repeated for each feature with missing values, with the newly imputed values from the previous step being used as input for the next.
5. The algorithm continues to iterate until the imputed values converge, meaning the changes between iterations are negligible.

Rationale for using IterativeImputer: The choice to use `IterativeImputer` was deliberate due to its ability to preserve the underlying relationships between variables. By iteratively predicting missing values, the model learns the inherent structure of the data. This approach avoids introducing biases or distorting the data's distribution, which can be a risk with simpler imputation methods. As a result, the imputed dataset is a more accurate and robust representation of the original data, which is essential for the subsequent analytical steps.

2.3 Exploratory Data Analysis (EDA)

To better understand the relationships among the GEM variables, several visualization tools were applied:

- **Line plots:** to display the temporal evolution of exports, imports, total reserves.

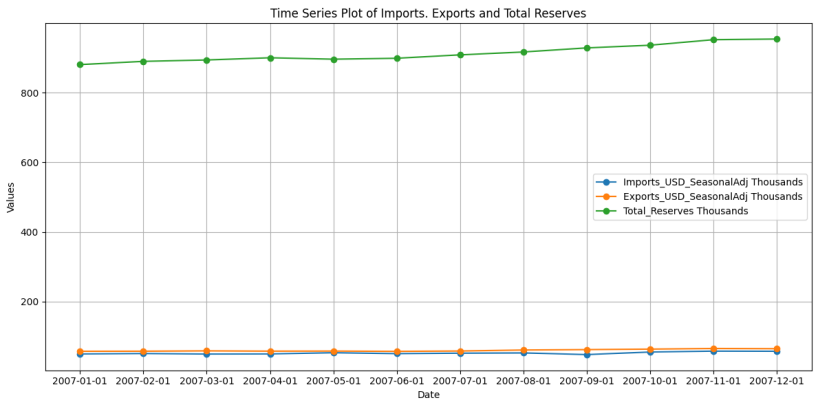

Figure 1: Time series for japan in 2007

- **Correlation matrix:** To further refine our feature set and address potential multicollinearity, a Correlation matrix was generated to visualize the correlations across all variables. This visual analysis provided clear insight into which features were highly correlated. The Correlation matrix served as a powerful tool to validate our feature selection process, ensuring that the final set of variables was not only robust but also non-redundant.

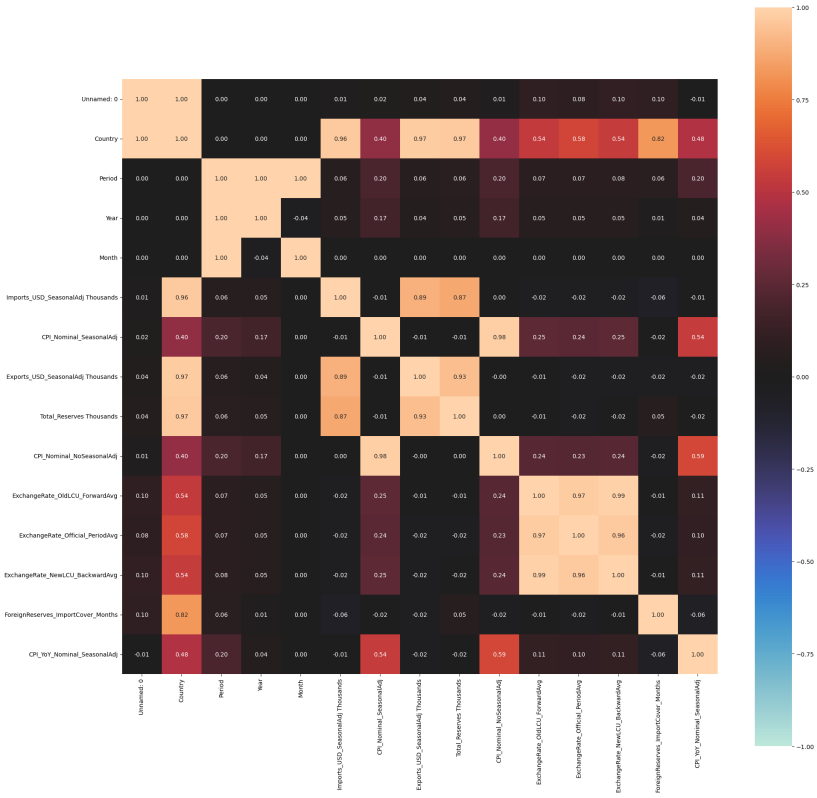

Figure 2: Correlation matrix

3 Clustering and Analysis

Clustering analysis was applied to group countries into homogeneous clusters based on their macroeconomic and trade indicators. The objective was to identify groups of economies with similar structural characteristics, so that representative countries from each cluster could be selected for the forecasting stage. Two algorithms were considered for this task: **K-Means** and **K-Medoids**. Both are partitional clustering methods but differ in how they define cluster centers.

3.1 Methodological Approach

K-Means defines the cluster center as the *mean* of all points in the cluster and assigns each country to the cluster with the nearest centroid. It minimizes the squared Euclidean distance between data points and their assigned center, making it computationally efficient and well-suited to large datasets. However, it is sensitive to outliers, as the mean can be distorted by extreme values.
K-Medoids, in contrast, selects an actual data point (the *medoid*) as the cluster center. This makes it more robust to outliers, since the medoid is less influenced by extreme observations. The trade-off is higher computational cost, which makes it less scalable for very large datasets.

3.2 Determining the Number of Clusters

The optimal number of clusters was evaluated using the **silhouette score**, which measures both cohesion (how close points are within the same cluster) and separation (how distinct clusters are from one another). The score ranges from -1 to $+1$, with higher values indicating more meaningful clustering structures.
The results indicated that a two-cluster solution ($K = 2$) produced the highest silhouette score (above 0.5), significantly outperforming other tested values. This suggests that the dataset can be meaningfully divided into two distinct groups of countries.

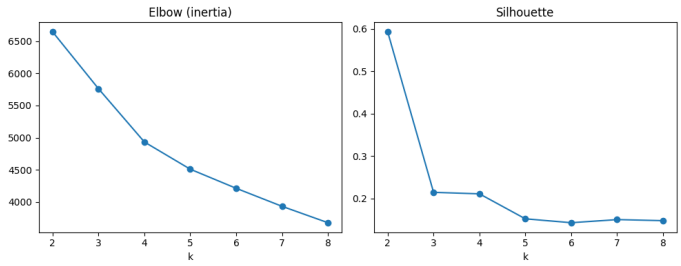

Figure 3: Silhouette score for different values of K

3.3 Comparison of K-Means and K-Medoids

To evaluate the consistency between the two algorithms, the **Adjusted Rand Index (ARI)** was computed. The ARI compares two clustering solutions on the same dataset, with values close to 1.0 indicating strong agreement and values near 0.0 indicating similarity no better than random chance.
In this case, the ARI score between K-Means and K-Medoids was 0.022, showing that the two methods produced materially different partitions. This reflects their different optimization criteria: K-Means captures average-based groupings, whereas K-Medoids emphasizes robustness to outliers. Given that K-Means provided clearer group averages and better supported the interpretability objective of this study, it was selected as the main clustering method.

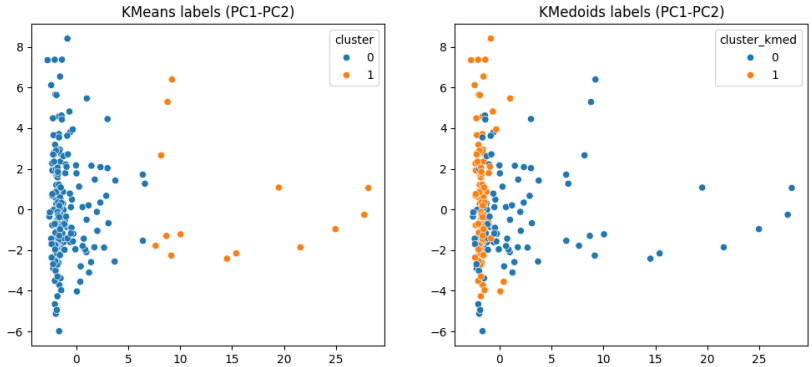

Figure 4: Scatterplot of clustering results from K-Means and K-Medoids

3.4 Interpretation

The final K-Means solution divided countries into a more seperated clusters that reflect structural differences in trade and macroeconomic indicators. From each cluster, two representative economies were selected for subsequent forecasting analysis. This approach ensures that the forecasting stage incorporates both dominant trade powers(cluster 1 - in orange) and under-developed/emerging economies(cluster 0 - in blue), allowing for more balanced insights into future global trade dynamics.

3.5 Hierarchical dendrogram

A dendogram is a tree-like diagram used to visualize the results of hierarchical clustering. It allows to visually inspect how the countries are related. We can see a main split between developed and developing nations, with further subdivisions within those groups.

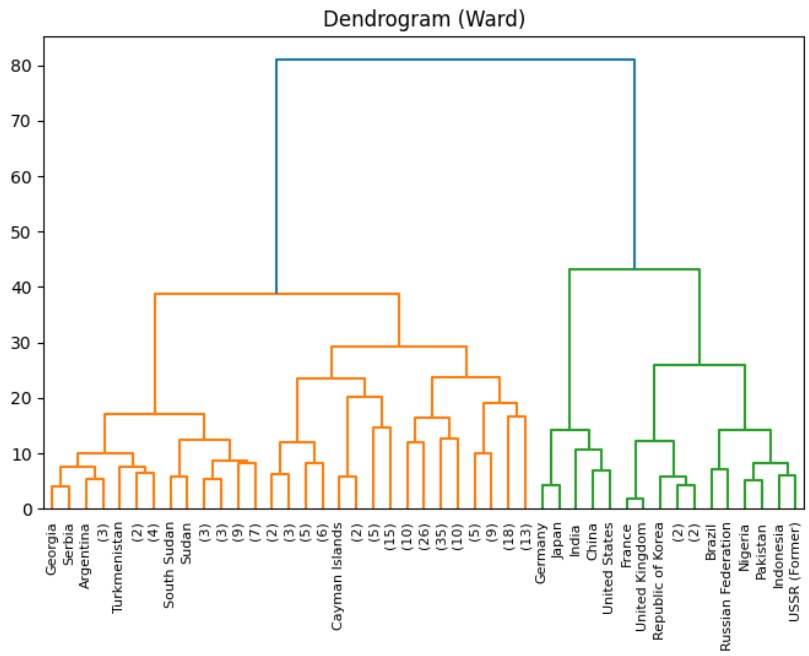

Figure 5: dendrogram

A world map illustrating the final clustering of countries. This visualization provides a geographical context to the clusters identified by the K-Means algorithm, showing the global distribution of the two distinct groups.

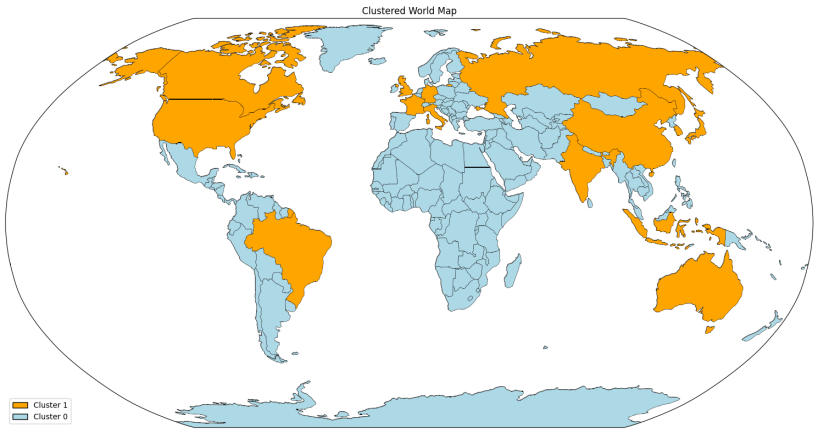

Figure 6: A world map illustrating the final clustering of countries

3.6 Selection of Representative Countries

After obtaining the two-cluster solution, we selected representative economies to carry forward into the forecasting stage. The choice of countries was guided by three main criteria:

- **Cluster representation:** At least one country was chosen from each cluster to ensure that both groups of economies were reflected in the forecasting stage.
- **Economic significance:** Priority was given to economies with large GDP, trade volumes, or demographic weight, in line with the feature importance analysis, which highlighted GDP, population, and trade shares as the most decisive variables in the clustering.
- **Data availability:** Countries with reliable and sufficiently long time series in the GEM dataset were selected to ensure robust forecasting.

Following these criteria, we selected:

- **Brazil (Cluster 1):** Brazil represents a large emerging economy with a significant role in global trade flows. Its GDP and population size were among the most important variables in the clustering process. Brazil's position as Latin America's largest economy makes it a strong representative of economies in Cluster 0, which grouped together advanced economies and large emerging markets.
- **Ghana (Cluster 0):** Ghana was selected as the representative of Cluster 0, which included smaller and less diversified economies. Ghana stands out within this group as Africa's largest economy by GDP and, with increasing importance in global energy exports. Including Ghana allows us to capture the dynamics of a rapidly growing but structurally different economy compared to Brazil.

This selection ensures that the forecasting stage reflects both the scale and diversification of a major emerging market (Brazil) and the demographic and resource-driven potential of a developing economy (Ghana). Together, they provide a balanced perspective on future trade resilience and competitiveness across clusters.

4 Models

To forecast future trade performance, two different time series forecasting models were deployed: **SARIMAX** and **Facebook Prophet**. Using both approaches allows us to compare traditional statistical modeling with more flexible machine learning techniques.

4.0.1 SARIMAX

To forecast trade-related indicators, we relied on the Seasonal AutoRegressive Integrated Moving Average with eXogenous factors (SARIMAX) model. SARIMAX is an extension of the classical ARIMA model that is particularly suitable for economic time series because it combines three important features:

- **Trend and integration:** it can handle data that evolve over time and are not stationary, meaning their average level changes.
- **Seasonality:** it explicitly models repeating patterns, such as monthly or yearly cycles in exports and imports.
- **Exogenous variables:** it allows the inclusion of additional predictors (such as CPI, reserves, or exchange rates), which can improve the forecast by linking trade flows to other macroeconomic drivers.

The general notation for a SARIMAX model is:

$$SARIMA(p,d,q) \times (P,D,Q)_s$$

where:

- p,d,q describe the non-seasonal part (autoregression, differencing, and moving average).
- P,D,Q describe the seasonal part of the model.
- s is the length of the seasonal cycle (for monthly data, $s = 12$).

This structure makes SARIMAX flexible enough to capture both short-term fluctuations and long-term seasonal effects. In our context, exports are influenced not only by global demand cycles but also by recurring patterns linked to production and consumption, which justifies the use of a seasonal specification. Compared to simpler forecasting methods (such as exponential smoothing or linear regression), SARIMAX has the advantage of capturing autocorrelation and seasonality in a systematic way. However, it is more complex to estimate and interpret, and results may be sensitive to parameter choices.

Outliers check in series: Outliers are extreme values that deviate significantly from normal patterns and can distort forecasting models. We use three approaches Z-score (statistical deviation), IQR (interquartile range), and Isolation Forest (machine learning) - to identify outliers with different sensitivities. Handling strategies implemented include complete removal, filling gaps with estimated values, median/mean, and winsorization. Based on outlier percentage, it suggests optimal treatment methods - interpolation for few outliers (<2%), rolling median for moderate cases (<5%), and winsorization for extensive outliers. Proper outlier treatment improves data quality, model accuracy, and ensures more reliable statistical inferences by removing or correcting data points that could mislead analysis while preserving the data structure and temporal relationships.

Series Transformation: Transformations stabilize variance over time, normalize skewed distributions, linearize exponential trends, and ensure model residuals meet statistical assumptions. This step improves forecasting accuracy, and ensures proper statistical inference, allowing time series models to perform optimally. For example Box-Cox transformation, which is often preferred for economic data adapts to the specific characteristics of the series, making it ideal for export/import data that typically shows exponential growth patterns and changing variance levels.

Grid Search to find optimal parameters: Manual parameter selection is time consuming so automated hyperparameter tuning was used for SARIMAX time series models. It tested different parameter combinations to find the optimal model configuration. ARIMA parameters (p, d, q) ranging from 0-1 were explored,

where p=autoregressive terms, d=differencing order, q=moving average terms. Seasonal parameters used the same ranges with a 12-month cycle, appropriate for monthly data with annual seasonality patterns. All possible parameter combinations were tested using AIC (Akaike Information Criterion) as the selection metric. The process ranks all valid models by AIC score, with lower values indicating better performance.

4.0.2 FACEBOOK PROPHET

Facebook's Prophet algorithm, a robust forecasting tool designed for business time series with strong seasonal patterns and trend changes. Prophet is effective for data with missing values, outliers, and multiple seasonal components.
The Prophet model is fitted to historical data and generates 24-month ahead forecasts with monthly frequency. Prophet automatically detects trend changes, seasonal patterns, and provides uncertainty intervals around predictions.
The forecast plot displays historical data, predicted values, and confidence bands, while the components plot decomposes the time series into trend, seasonal, and residual components. This decomposition reveals underlying patterns like annual seasonality and long-term trends. Prophet is a decomposable time-series model developed by Meta, defined as:

$$y(t) = g(t) + s(t) + h(t) + \varepsilon_t,$$

where $g(t)$ models trend, $s(t)$ captures seasonality, $h(t)$ incorporates holiday effects, and ε_t is the error term.
Pros: Handles multiple seasonalities, robust to missing data and outliers, intuitive parameterization.
Cons: Less precise for very short time horizons; limited interpretability compared to ARIMA.

4.1 Model Selection and Evaluation

Models were trained on historical data (2006–2020) and validated on more recent years (2021–2024). Forecast accuracy was measured using common error metrics such as Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). This ensured that the chosen models were both accurate and generalizable.

5 Experiments

5.1 Parameter Selection and Model Estimation

To identify the most suitable SARIMAX specification, a grid search procedure was applied over a set of autoregressive (AR), moving average (MA), and seasonal parameters. A total of 64 candidate models were tested, with the best-performing specification selected according to the Akaike Information Criterion (AIC). The top 5 models found initial had the following parameters:

```
Total combinations: 64
SARIMAX parameters search: 100%|██████████| 8/8 [00:26<00:00,  3.26s/it]
Best model found:
order = (1, 1, 0)
seasonal_order = (0, 0, 0, 12)
AIC = -532.5770828847017

Top 5 modelli:
       order seasonal_order         AIC
0  (1, 1, 0)  (0, 0, 0, 12) -532.577083
1  (1, 0, 1)  (0, 0, 0, 12) -531.739293
2  (0, 1, 1)  (0, 0, 0, 12) -530.782914
3  (1, 1, 1)  (0, 0, 0, 12) -529.332662
4  (1, 0, 0)  (0, 0, 0, 12) -517.968031
```

Figure 7: Top 5 models From Grid Search

which outperformed alternative specifications. This configuration indicates the presence of both autoregressive and moving-average dynamics, as well as a seasonal component with a yearly frequency.

5.2 Model Diagnostics

To validate the adequacy of the SARIMAX model, residual diagnostics were conducted. Figure 8 reports the standardized residuals, their distribution, Q-Q plot, and correlogram:

- The residuals fluctuate randomly around zero, suggesting the absence of systematic patterns.
- The correlogram shows no significant autocorrelation, indicating that most of the temporal dependence was captured by the model.
- The Q-Q plot highlights some deviations from normality in the tails, consistent with the Jarque–Bera test.

Despite minor deviations from normality, the model provides an adequate representation of the underlying process.

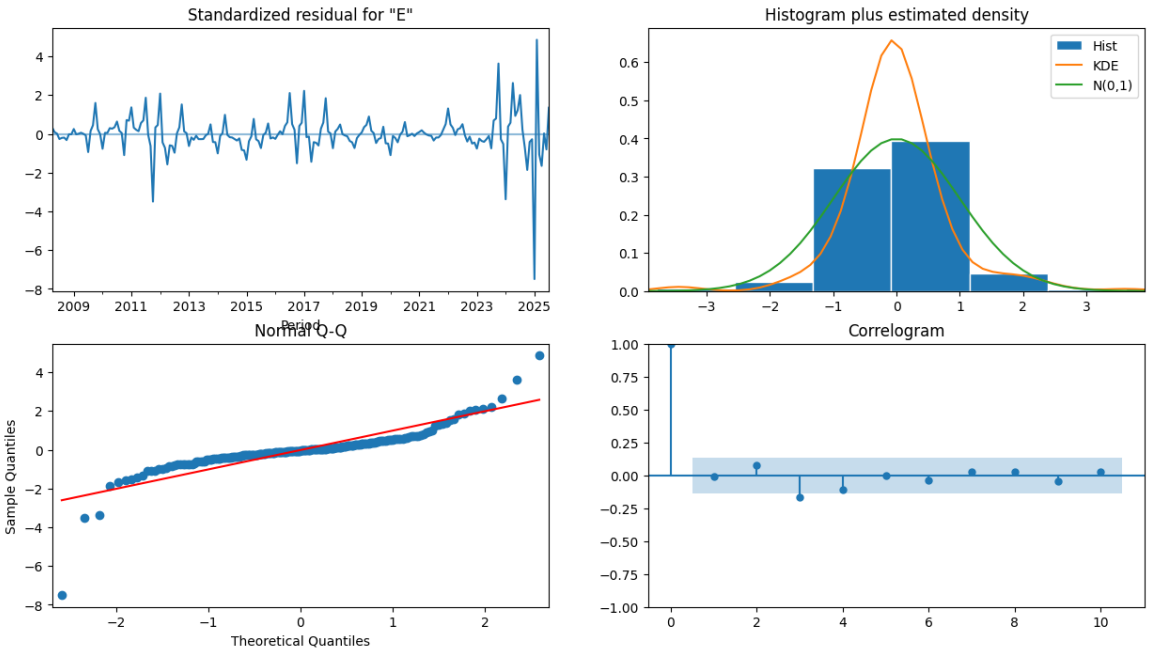

Figure 8: Residual diagnostics for SARIMAX model

5.3 Outlier Detection and Treatment

Given the importance of robust forecasting, and given the model diagnostics showed previously which has some deviations from normality in the tails (Q-Q plot), potential outliers in the export series were also investigated. Figure 9 illustrates the detection of anomalies through multiple approaches: isolation-based methods, boxplot visualization, histogram inspection, and Z-scores over time.
Several anomalous values were identified, particularly during early 2010s and recent years, possibly linked to global financial instability and post-pandemic trade shocks. After identification, these extreme values were adjusted to reduce their undue influence on parameter estimation. The corrected series (post-elimination) was subsequently used for the final forecasting stage.

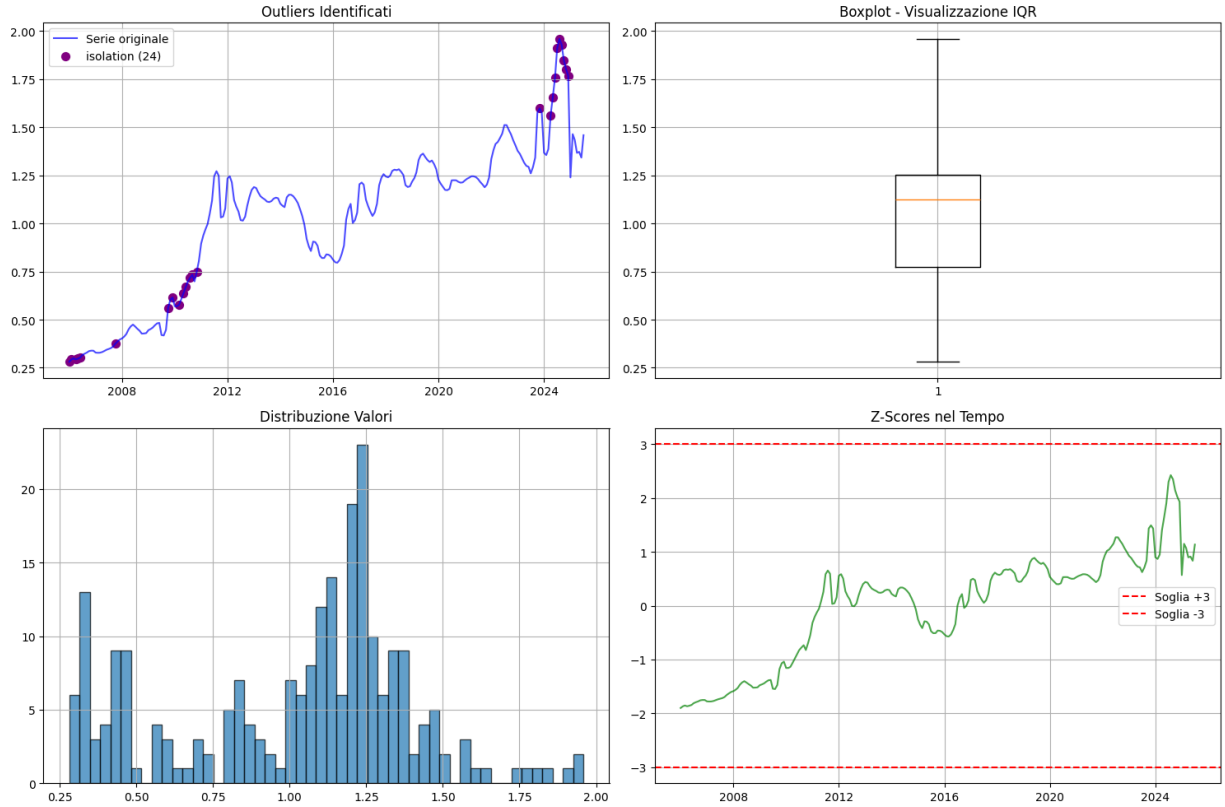

Figure 9: Example of Detection of outliers in the Ghana export time series

We then apply a Box-Cox transformation to stabilize variance over time, normalize skewed distributions, linearize exponential trends, and ensure model residuals meet statistical assumptions. After pre-processing our data we re-do the grid search and re-fit our model with the following parameters:

$$\text{order} = (1, 1, 1), \quad \text{seasonal order} = (0, 1, 1, 12),$$

yielding an AIC of -537.04.

6 Results

6.1 Ghana forecast

The SARIMAX model for Ghana projects continued volatility in exports, with a predicted median value fluctuating between 1.25 and 1.5 billion USD. The forecast is accompanied by a wide 95% confidence interval, indicating significant model uncertainty. A Monte Carlo simulation based on the SARIMAX parameters provides a more stable outlook, with a median prediction of approximately 1.5 billion USD and a tighter confidence interval. In contrast, the Prophet model forecasts a clear and sustained upward trend in Ghana's exports, underpinned by a distinct yearly seasonal pattern. This model projects export values to rise steadily from approximately 1.5 billion USD in 2025 to around 1.7 billion USD by the end of 2027.

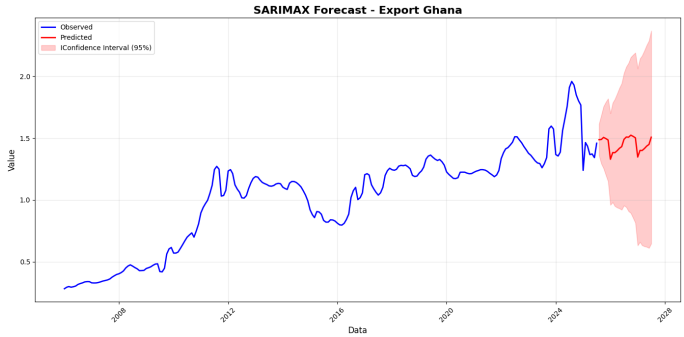

Figure 10: SARIMAX Model Fit and Historical Data for Ghana

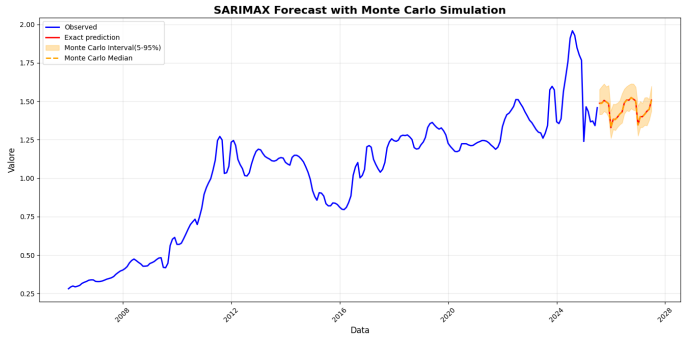

Figure 11: SARIMAX Forecast with Monte Carlo Simulation for Ghana

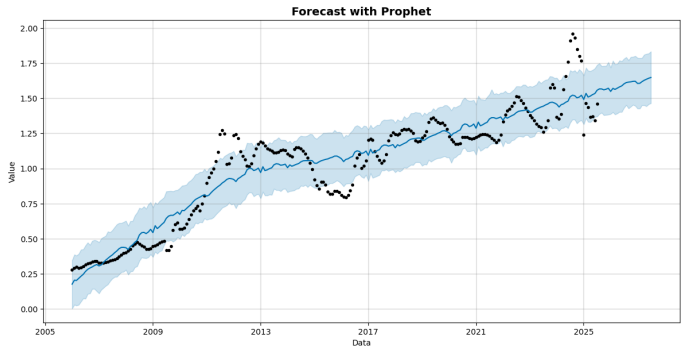

Figure 12: Prophet Forecast for Ghana

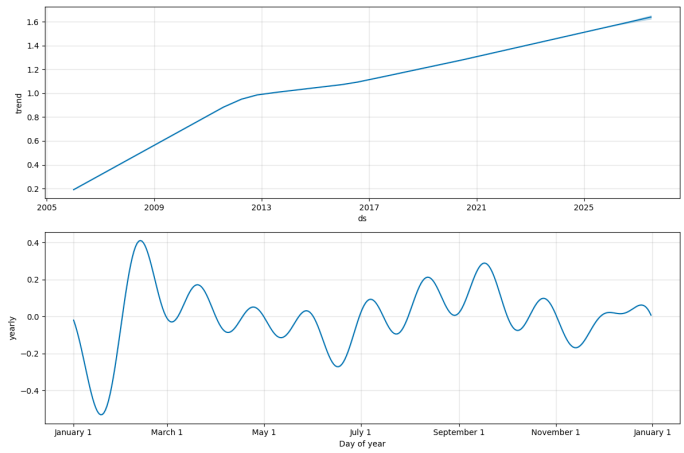

Figure 13: Ghana Export Forecast with Confidence Intervals

6.2 Brazil forecast

For Brazil, the SARIMAX model forecasts a sharp decline in export values from recent highs, stabilizing at a lower range of 15 to 17 billion USD. The accompanying 95% confidence interval is exceptionally wide, reflecting the high volatility of Brazil's historical trade data. The Monte Carlo simulation supports this forecast of a significant decline but offers a more constrained range of potential outcomes. The Prophet model, however, presents a more optimistic outlook, forecasting a gradual but consistent increase in exports from a 2025 low, driven by a positive long-term trend. The model predicts a rise in export values from roughly 25 billion USD to over 30 billion USD by 2028.

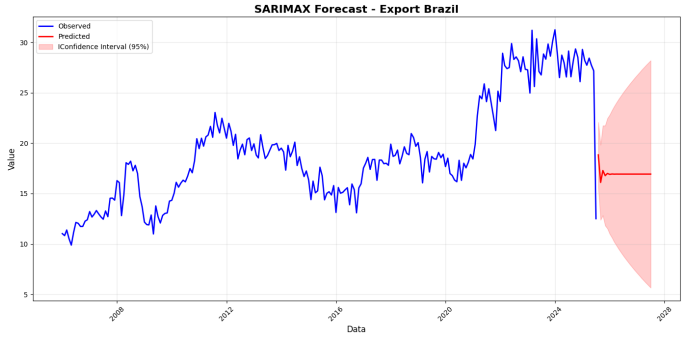

Figure 14: SARIMAX Forecast for Brazil

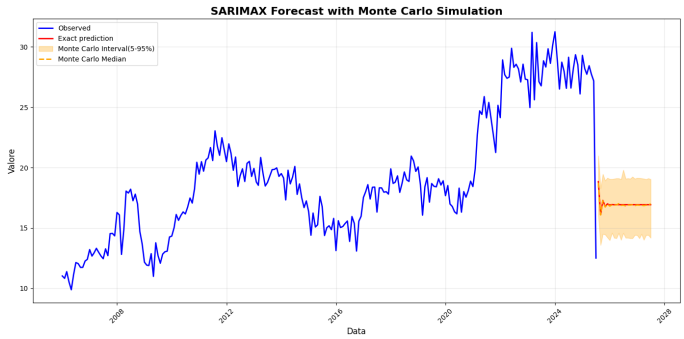

Figure 15: SARIMAX Forecast with Monte Carlo Simulation for Brazil

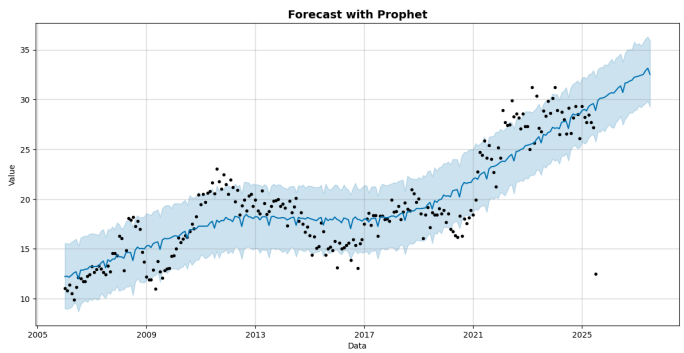

Figure 16: Prophet Forecast for Brazil

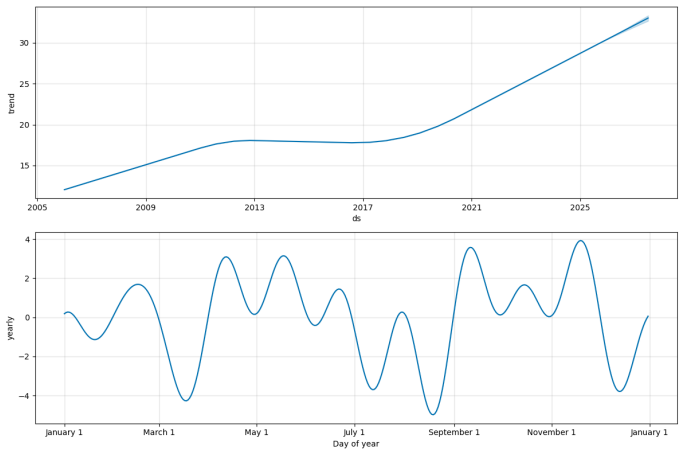

Figure 17: Prophet Forecast Components for Brazil Exports

7 Conclusions

The analysis reveals two diverging narratives for future trade dominance, highlighting a fundamental difference between the selected economies.

For Ghana, the findings point to a stable and resilient upward trajectory. The strong seasonal component identified by the Prophet model suggests a consistent, cyclical rhythm in its trade economy, while the long-term trend points toward continued growth. This resilience indicates that Ghana, as a representative of its cluster, is on a path to strengthen its position in global trade, driven by predictable patterns and a positive trend.

In contrast, the outlook for Brazil is characterized by significant uncertainty and potential for a sharp decline. While the Prophet model projects sustained growth, the SARIMAX model, which better captures recent extreme fluctuations, forecasts a major downturn. This divergence underscores the high volatility inherent in Brazil's economy. The ability of the economy to absorb shocks and sustain its trade position is therefore a key determinant of its future dominance.

In summary, the forecasting results suggest that while emerging economies like Ghana demonstrate a clear path toward sustained growth and trade resilience, larger economies like Brazil face a more complex and uncertain future, heavily influenced by periods of extreme volatility. This dichotomy reflects the shifting balance of global trade, where traditional powerhouses may face challenges while smaller, structurally distinct economies steadily increase their influence.

7.1 Future Improvements

To build on these findings and enhance the robustness of the analysis, several improvements could be pursued in future work:

1. Advanced Modeling: The forecasting could be improved by exploring more sophisticated models. While SARIMAX and Prophet are effective, future research could leverage ensemble models or deep learning architectures like Long Short-Term Memory (LSTM) networks to capture more complex, non-linear relationships in the data.

2. Expanded Feature Set: The current models could be strengthened by incorporating additional external data. Integrating variables such as global commodity prices (e.g., for oil, cocoa, or iron ore), global GDP growth rates, and relevant trade policy indices would provide a more complete picture of the drivers of trade dominance.

References

[1] Global economy indicators, 2023.

[2] United Nations Statistics Division. United nations system of national accounts and annual national accounts database (sna / ama), 2025.

[3] The World Bank. Global economic monitor (gem), 2025.

A Appendix

Table 5: Complete list of constructed features: definition and interpretation

Feature	Definition / formula	Economic interpretation
population_latest	Latest available population level	Captures demographic scale of the country.
gdp_latest	Latest available GDP level	Absolute size of the economy.
gni_percap_latest	GNI (or GDP) divided by population	Standard proxy for economic development.
exports_share_gdp	mean(Exports)/mean(GDP)	Export dependence.
imports_share_gdp	mean(Imports)/mean(GDP)	Import dependence.
trade_openness	exports_share_gdp + imports_share_gdp	Overall trade integration.
exp_mean_growth	mean($\Delta\log(Exports_t)$)	Average export growth rate.
exp_vol_growth	std($\Delta\log(Exports_t)$)	Export growth volatility.
exp_ac1	Autocorr lag 1 of $\Delta\log(Exports)$	Persistence of export dynamics.
exp_ac12	Autocorr lag 12 of $\Delta\log(Exports)$	Seasonality in exports.
exp_down_share	Share of periods with $\Delta\log(Exports) < 0$	Frequency of export contractions.
exp_mean_log	mean($\log Exports_t$)	Average log-level of exports.
exp_season_amp	Amplitude of seasonal component (STL)	Strength of export seasonality.
exp_max_drawdown	$\min_t(\log Exports_t - \max_{s\leq t}\log Exports_s)$	Largest cumulative export fall.
imp_mean_growth	mean($\Delta\log(Imports_t)$)	Average import growth.
imp_vol_growth	std($\Delta\log(Imports_t)$)	Import volatility.
gdp_mean_growth	mean($\Delta\log(GDP_t)$)	Average GDP growth.
gdp_vol_growth	std($\Delta\log(GDP_t)$)	GDP volatility.
gdp_max_drawdown	As above, applied to GDP	Maximum recession depth.
va_mean_growth	mean($\Delta\log(ValueAdded_t)$)	Growth of total value added.
va_vol_growth	std($\Delta\log(ValueAdded_t)$)	Volatility of value added.
corr_exp_gdp	corr($\Delta\log Exports$, $\Delta\log GDP$)	Coupling of external and domestic activity.
inv_share_gdp	mean(Inventory change)/mean(GDP)	Volatile demand component.
household_cons_share_gdp	mean(Household consumption)/mean(GDP)	Structure of private demand.
gov_cons_share_gdp	mean(Government consumption)/mean(GDP)	Role of public demand.
manufacturing_share_va	mean(VA_{manuf}/VA_{total})	Industrialization degree.
agri_share_va	mean(VA_{agri}/VA_{total})	Agricultural dependence.
mining_manuf_utils_share_va	mean(VA_{C-E}/VA_{total})	Commodity/industry weight.
services_trade_share_va	mean(VA_{G-H}/VA_{total})	Services/tourism share.
transport_share_va	mean(VA_I/VA_{total})	Logistics and communications.
primary_share	Combined agriculture+mining share	Weight of primary sectors.
sector_HHI	$\sum_i s_i^2$, with s_i =sector share	Sectoral concentration / diversification.
fx_mean	mean(log FX level)	Average exchange rate level.
fx_vol	std($\Delta\log(FX)$)	Exchange rate volatility.
fx_trend	slope of $\log(FX_t)$	Long-term depreciation/appreciation.
fx_ac1	Autocorr lag 1 of $\Delta\log(FX)$	Persistence of FX changes.
exp_skew	skewness of $\log(Exports)$	Asymmetry of export shocks.

Feature	Definition / formula	Economic interpretation
exp_kurt	kurtosis of $\log(Exports)$	Tail risk of export shocks.
unemp_mean (if available)	mean unemployment rate	Labor market/social indicator.
real_exp_mean_growth (if CPI available)	log(exports) – inflation	Real export growth.
exp_deflated_trend (if CPI available)	slope(log exports – log CPI)	Long-run real export trend.
corr_exp_infl (if CPI available)	corr(log exports, inflation)	Link exports–inflation.
pass_through_expprice (if export price index available)	slope(log export price on inflation)	Price pass-through estimate.